REPORT 04 - Waterfall Plan

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CSE272 - Software Lifecycle Models

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1 OVERVIEW

Bank Associate Nation Keeper, Inc (B.A.N.K., Inc) has hired our company for building a mobile application that will help homeowners stay on top of their home maintenance. The mobile application includes features as follows:

- Monitors electricity usage,
- Monitor appliance upkeep,
- Monitor yard maintenance,
- Track periodic cleaning.

The application will provide a monthly checklist of home maintenance activities, allow the homeowner to check off items when they are complete, and produce a report at the end of the month.

In the development of this application, our company will implement the software development methodology proposed by Dr. Winston W. Royce (1970), commonly known as “Waterfall Model” (Dischave, 2012). The overall project duration has been estimated at 17 months.

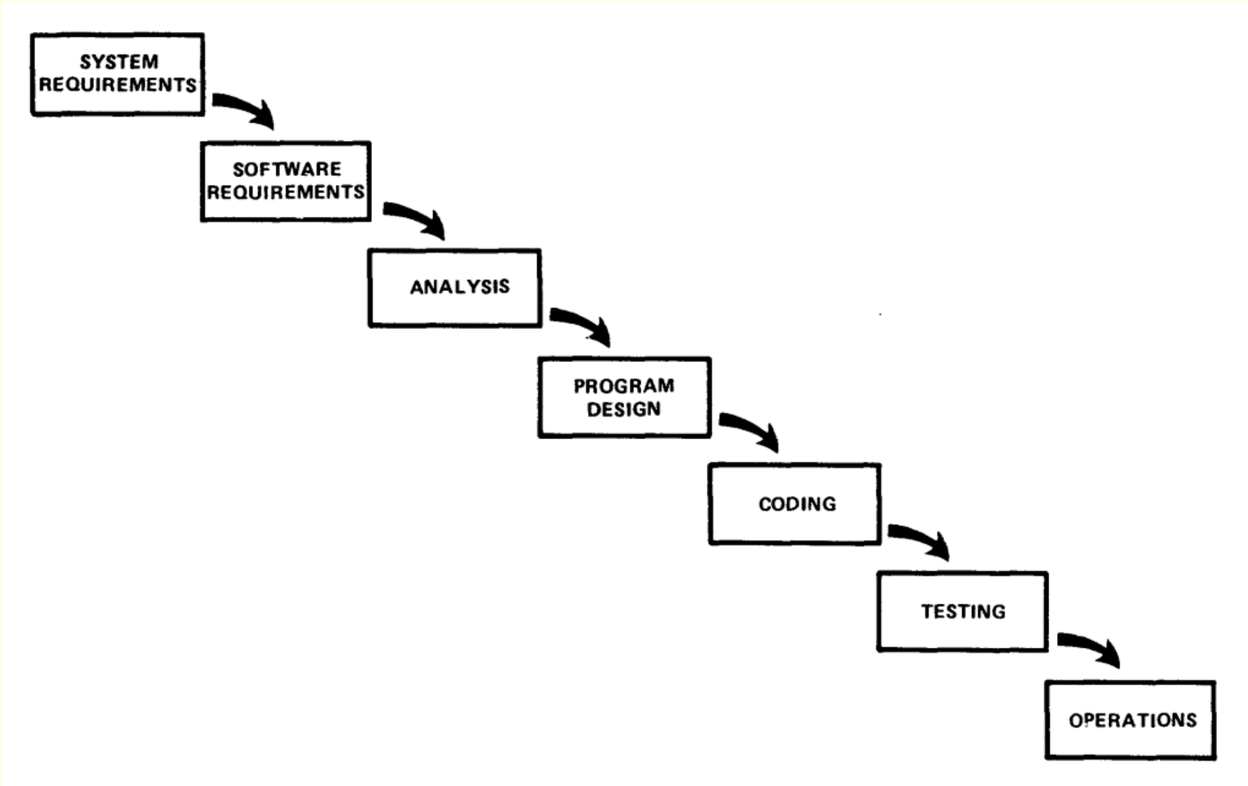

Figure 1 *- The so-called "Waterfall Model" in its simplest implementation.*

This model was proposed as a solution for the management of large software systems (Royce, 1970), and it should suffice the managemental requirements we need for the completion of this project. Furthermore, the overall model allows the avoidance of development risks by introducing five additional features (or phases):

1. *System Requirements Phase* and *Preliminary Program Design*: Emphasis on extensive design of the program before beginning analysis and coding.
2. *Documentation*: Complete documentation during all phases (except during the *System Requirements* and *Analysis* phases), especially during the design phase.
3. Do it twice by means of implementing a pilot model of the systems.
4. *Testing*: Meticulously plan, control and monitor testing,
5. Involve the customer.

These additional features increase the complexity of the system, as shown in Figure 2.

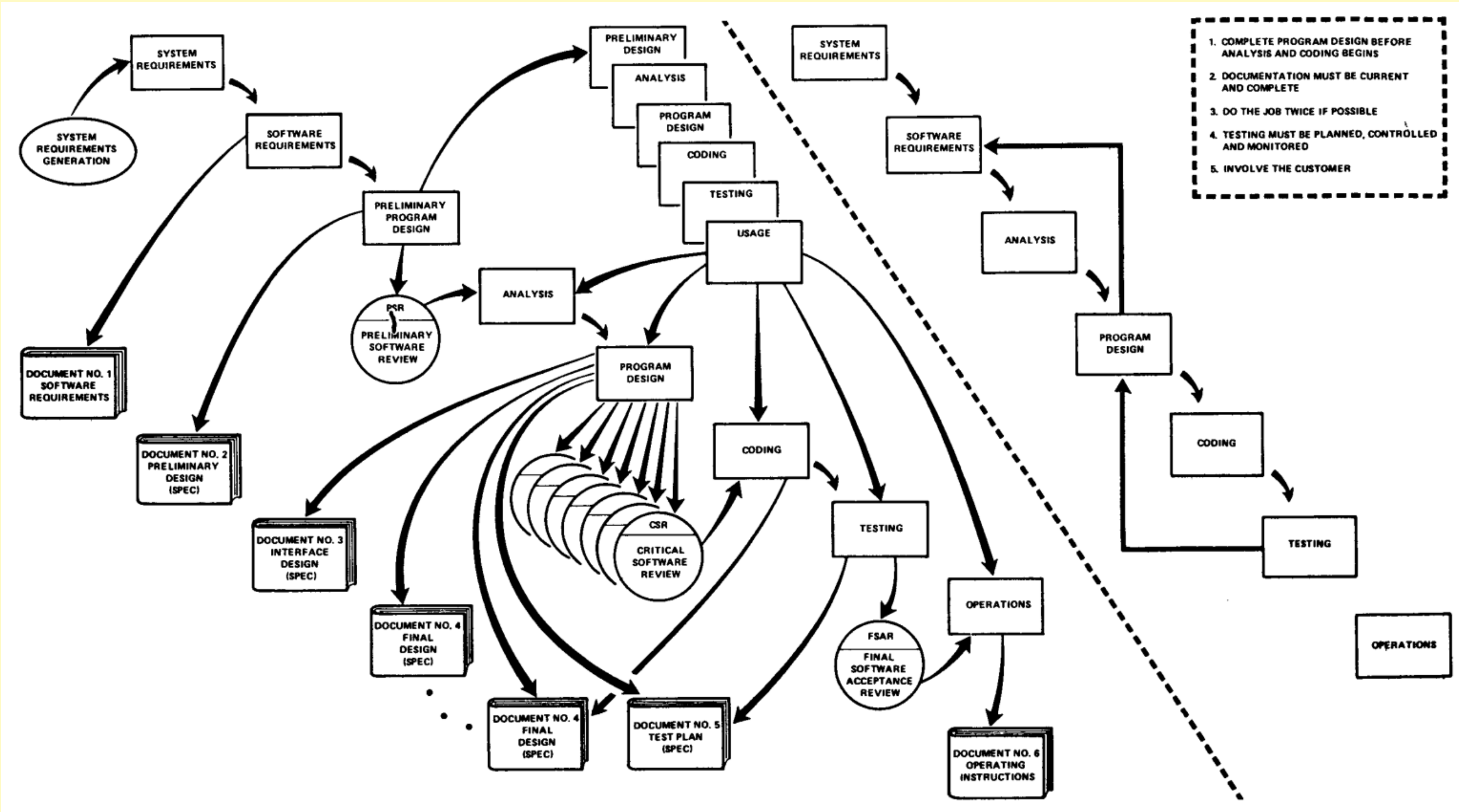

Figure 2 *- "Waterfall model" complete implementation.*

Representatives from B.A.N.K., Inc, Rick and Rachel are available for us. to address any questions and expectations. The following sections will describe the planning of: Meetings, Documents, Roles and Checkpoints.

Keywords: Royce, Waterfall Model, software development methodologies

2 MEETINGS

2.1. Introductory Meeting

2.1.1. Who

- *All employees, B.A.N.K. representatives Rick and Rachel.*

2.1.2. Agenda

- *Welcome*
- *Overview and detailed explanation of the software to be developed.*
- *Role assignment and explanation.*
- *Question and answer session.*

2.1.3. Purpose

- *Kick-off the project and build excitement for the task at hand! Assign employees to teams, appoint team leads, and allow employees the opportunity to meet with B.A.N.K. representatives.*

2.1.4. Event

- *Held once at the beginning of the project.*

2.2. Team Meetings (start of week)

2.2.1. Who

- *Coding and Testing Team Members, in respective groups. Other employees may be invited depending on need.*

2.2.2. Agenda

- *Welcome*
- *Overview of week's tasks. Check for pending tasks.*
- *Assignment of tasks.*

- *Questions and answer session with Team Lead.*

2.2.3. Purpose

- *Assign weekly tasks to employees, and, if applicable, check for pending tasks to be assigned in coming weeks. Allow employees the opportunity to discuss concerns with the team lead.*

2.2.4. Event

- *This meeting will occur at the beginning of every week.*

2.3. Team Meeting (end of week)

2.3.1. Who

- *Coding and Testing Team Members, in respective groups. Other employees may be invited depending on need.*

2.3.2. Agenda

- *Welcome*
- *Week's work review.*
- *Problem solving discussion.*
- *Mapping of next week's TODOs.*

2.3.3. Purpose

- *Discuss the week's successes and failures. Research solutions to problems encountered. Identify the next tasks that need to be done, and make a plan for the next week.*

2.3.4. Event

- *This meeting will occur at the end of each week.*

2.4. Checkpoint Meetings

2.4.1. Who

- *Coding and Testing Team Members, in respective groups. Other employees may be invited depending on need.*

2.4.2. Agenda

- *Welcome*
- *Overview of finished module*
- *Discussion of next steps/functionality to work on.*
- *Introduction of new checkpoint*
- *Distribution of new tasks on teams.*

2.4.3. Purpose

- *Each time a checkpoint is reached this meeting will be held.*
- *During this meeting, everything will be discussed pertaining to the work already accomplished and the work that needs to be done.*
- *After new tasks are identified for the next level of functionality to be developed, those tasks are assigned to the different teams.*
- *This is an overview and summary of one checkpoint and the kickoff of the next.*

2.4.4. Event

- *This meeting will occur upon completion of a checkpoint.*

2.5 System Requirements Meeting

2.5.1. ***Who***

- *Coding Team Lead, Project Manager, B.A.N.K representatives Rick and Rachel*

2.5.2. ***Agenda***

- *Welcome*
- *Discussion of necessary system requirements*

2.5.3. ***Purpose***

- *Define system requirements for the development and functionality of the product.*

2.5.4 ***Event***

- *This meeting will occur once upon the beginning of the System Requirements phase - approximately Day Two of the project.*

2.6 Software Requirements Meeting

2.6.1. ***Who***

- *Coding Team Lead, Testing Team Lead, Project Manager, B.A.N.K representatives Rick and Rachel*

2.6.2. ***Agenda***

- *Welcome*
- *Discussion of necessary software requirements*

2.6.3. ***Purpose***

- *Define software requirements for the development and functionality of the product.*

2.6.4 ***Event***

- *This meeting will occur once upon the beginning of the Software Requirements phase - approximately week 2 of the project.*

2.7 System Requirements Meeting

2.7.1. ***Who***

- *Coding Team Lead, Project Manager, B.A.N.K representatives Rick and Rachel*

2.7.2. ***Agenda***

- *Welcome*
- *Discussion of necessary system requirements*

2.7.3. ***Purpose***

- *Define system requirements for the development and functionality of the product.*

2.7.4 ***Event***

- *This meeting will occur upon the beginning of the System Requirements phase - approximately Day Two of the project.*

2.8 Preliminary Design Meeting

2.8.1. ***Who***

- *Coding Team Lead, Project Manager, B.A.N.K representatives Rick and Rachel, Technical Writer*

2.8.2. ***Agenda***

- *Welcome*
- *Discussion of preliminary design*
- *Outline the Preliminary Design Specification Document*

2.8.3. ***Purpose***

- *Discuss preliminary design requirements with the assistance of client representatives. Initialize creation of the Preliminary Design Specification Document with the help of the Technical Writer.*

2.8.4 ***Event***

- *This meeting will occur upon the beginning of the Preliminary Program Design phase - approximately 1.5 months after the project start date.*

2.9 Preliminary Software Review (PSR)

2.9.1. ***Who***

- *Project Manager, B.A.N.K representatives Rick, Rachel, Pat, and Patricia, Technical Writer*

2.9.2. ***Agenda***

- *Welcome*
- *Client review of Software Requirements Document*
- *Client review of Preliminary Design Document*

2.9.3. ***Purpose***

- *Client representatives will review Software Requirements and Preliminary Design documents. Feedback and suggested changes will be supplied.*

2.9.4 ***Event***

- *This meeting will occur once upon the end of the Preliminary Software Review phase - approximately 2.5 months after project start.*

2.10 Critical Software Review (CSR) Meeting #1

2.10.1. ***Who***

- *Project Manager, B.A.N.K representatives Rick, Rachel, Pat, and Patricia, Coding Team Lead, Technical Writer*

2.10.2. ***Agenda***

- *Welcome*
- *Review the first draft of the Final Design document*
- *Review the Interface Design document*

2.10.3. ***Purpose***

- *Client representatives will review the first draft of the Final Design document and the Interface Design document. Feedback and suggested changes will be supplied.*

2.10.4 ***Event***

- *This meeting will occur once during the Program Design phase - approximately 4 months after project start.*

2.11 Critical Software Review (CSR) Meeting #2

2.11.1. ***Who***

- *Project Manager, B.A.N.K representatives Rick, Rachel, Pat, and Patricia, Technical Writer*

2.11.2. ***Agenda***

- *Welcome*
- *Review the revised draft of the Final Design document*

2.11.3. ***Purpose***

- *Client representatives will review the revised draft of the Final Design document. Feedback and suggested changes will be supplied.*

2.11.4 ***Event***

- *This meeting will occur once during the Program Design phase - approximately 4.5 months after project start.*

2.12 Test Plan Design Meeting

2.12.1. ***Who***

- *Project Manager, Testing Team Lead, Coding Team Lead, Technical Writer*

2.12.2. ***Agenda***

- *Welcome*

- *Outline a draft for the Test Plan document.*

2.12.3. ***Purpose***

- *After revision of the Final Design, a Test Plan Specifications document must be completed. This meeting will outline a draft for the Test Plan Document.*

2.12.4 ***Event***

- *This meeting will occur once during the Program Design phase - approximately 4.5 months after project start.*

2.13 Test Plan Design Meeting

2.13.1. ***Who***

- *Project Manager, Testing Team Lead, Coding Team Lead, Technical Writer*

2.13.2. ***Agenda***

- *Welcome*
- *Outline a draft for the Test Plan document.*

2.13.3. ***Purpose***

- *After revision of the Final Design, a Test Plan Specifications document must be completed. This meeting will outline a draft for the Test Plan Document.*

2.13.4 ***Event***

- *This meeting will occur once during the Program Design phase - approximately 4.5 months after project start.*

2.14 Final Design Specifications Meeting

2.14.1. ***Who***

- *Project Manager, Testing Team Lead, Technical Writer*

2.14.2. ***Agenda***

- *Welcome*
- *Outline a draft for the Test Plan document.*

2.14.3. ***Purpose***

- *After revision of the Final Design, a Test Plan Specifications document must be completed. This meeting will outline a draft for the Test Plan Specifications document and prepare it for use.*

2.14.4 ***Event***

- *This meeting will occur once during the Coding phase - approximately 13 months after project start.*

2.13 Final Acceptance Review (FSAR)

2.13.1. ***Who***

- *Project Manager, Testing Team Lead, Coding Team Lead, Technical Writer, UX Designers, B.A.N.K. Representatives Pat, Patricia, Rick, and Rachel.*

2.13.2. ***Agenda***

- *Welcome*
- *Client inspection of all documentation.*
- *Client inspection of all software.*

2.13.3. ***Purpose***

- *This is arguably the most important meeting in this development life cycle. The client representatives will thoroughly inspect, test, and review the completed software and documentation. All documents and software will be individually reviewed by the client. Any necessary changes will be recommended.*

2.13.4 ***Event***

- *This meeting will occur at least once during the end of the Testing phase - approximately 16 months after project start. It may need to be repeated until all documentation and software have been approved by the client.*

2.13 Operations and Project End

2.13.1. ***Who***

- *All employees, B.A.N.K. Representatives Pat, Patricia, Rick, and Rachel.*

2.13.2. ***Agenda***

- *Welcome*
- *Project overview and review*
- *Presentation of Operating Instructions*
- *Concluding remarks*

2.13.3. ***Purpose***

- *The software will be released as a finished product. This meeting will allow for a reflection on successes and failures encountered in the development of the project. The Operating Instructions will be presented.*

2.13.4 ***Event***

- *This meeting will occur once during the Operations phase, approximately 17 months after project start.*

3 DOCUMENTS

3.1. Overview Document

3.1.1. Author

- Technical Writer, Project Manager, Coding Team Lead, Testing Team Lead

3.1.2. Intended Audience

- Coding and Testing Team Members, UX Designers, product clientele

3.1.3. Purpose

- **An Overview document must be written to allow every employee an "elemental understanding of the system", and it must be "understandable, informative, and current". (Royce, 1970).** This document will allow clientele, developers, and testers to understand the project at a glance.

3.1.4. Deadline

- *2 weeks after project start.*

3.2. System Requirements

3.2.1. Author

- Technical Writer, Project Manager, Coding Team Lead

3.2.2. Intended Audience

- Coding and Testing Team Members, UX Designers, product clientele

3.2.3. Purpose

- *Royce's model calls for creation of a Software Requirements document. In the spirit of creating complete, effective documentation, a System Requirements document will also be produced. This document will outline hardware and system requirements necessary for the software to be developed and effectively* *run.* **This document will be created during the System Requirements phase of development, as per Royce's model (Royce, 1970).**

3.2.4. Deadline

- *1 month after the project starts.*

3.3. Software Requirements (Specifications)

3.3.1. Author

- Technical Writer, Project Manager, Coding Team Lead, Testing Team Lead

3.3.2. Intended Audience

- Coding and Testing Team Members, product clientele

3.3.3. Purpose

- The **Software Requirements Specification**, according to the IEEE Computer Society, *"is a detailed description of the intended purpose, features, functionality and other aspects of a software application" (IEEE Computer Society).* **This document will be created during the Software Requirements phase of development, as per Royce's model (Royce, 1970).** The SRS describes the functionality of the software, and guides the coding team towards creating a functional program.

3.3.4. Deadline

- *1.5 months after project start.*

3.4. Preliminary Design (Specifications)

3.4.1. Author

- Technical Writer, Project Manager, B.A.N.K. representatives Peter and Patricia

3.4.2. Intended Audience

- Coding and Testing Team Members, UX Designers, product clientele

3.4.3. Purpose

- The **Preliminary Design specification** will serve as an initial, prototypical design of the project. *It will clearly define each component's requirements and give an outline of features to meet these requirements (Purdue, 2024).* **This document will be created during the Preliminary Program Design phase of development, as per Royce's model (Royce, 1970).**

3.4.4. Deadline

- *1.5 months after project start.*

3.5. Interface Design (Specifications)

3.5.1. Author

- Technical Writer, Project Manager, B.A.N.K. representatives Peter and Patricia, UX Designers

3.5.2. Intended Audience

- UX Designers, Coding Team Members

3.5.3. Purpose

- The **Interface Design specification** will *define the different interfaces of the product and their functionalities (Arnab, 2015). It will contain images and references that can be utilized to design user interfaces in the project.* **This document will be created during the Program Design phase of development, as per Royce's model (Royce, 1970).**

3.5.4. Deadline

- *3.5 months after project start.*

3.6. Final Design First Draft

3.6.1. Author

- Technical Writer, Project Manager, B.A.N.K. representatives Peter and Patricia, Coding Team Lead

3.6.2. Intended Audience

- Coding and Testing Team Members, product clientele

3.6.3. Purpose

- The **Final Design First Draft is an updated version of the Preliminary Design document created earlier. It will serve as a more complete version of the Preliminary Design document.** It will be superseded by the Revised Draft, which will be approved by B.A.N.K. executives. **This document will be created during the Program Design phase of development, as per Royce's model (Royce, 1970).**

3.6.4. Deadline

- *4 months after the project starts.*

3.7. Final Design Revised Draft

3.7.1. Author

- Technical Writer, Project Manager, B.A.N.K. representatives Peter and Patricia, Coding Team Lead

3.7.2. Intended Audience

- Coding and Testing Team Members, product clientele

3.7.3. Purpose

- **The Final Design First Draft will be revised after the first CSR meeting.** This document will be an updated version of the Final Design First Draft that more closely aligns with client expectations. **This document will be created during the Program Design phase of development, as per Royce's model (Royce, 1970).**

3.7.4. Deadline

- *4.5 months after project start.*

3.8. Final Design (As Built)

3.8.1. Author

- Technical Writer, Project Manager, B.A.N.K. representatives Peter and Patricia, Coding Team Lead

3.8.2. Intended Audience

- Coding and Testing Team Members, product clientele, future Operations engineers

3.8.3. Purpose

- **This is the completed version of the Final Design specifications document.** It will be updated after the Coding phase to become a comprehensive guide to the software. **This document will be created during the beginning of the Testing phase of development, as per Royce's model (Royce, 1970).**

3.8.4. Deadline

- *12.5 months after project start.*

3.9. Test Plan Specification

3.9.1. Author

- Technical Writer, Project Manager, Testing Team Lead

3.9.2. Intended Audience

- Testing Team Members

3.9.3. Purpose

- The **Testing Plan specifications document** will outline all tests to be performed on the developed software. *It will summarize all scenarios to be tested, describe how they will be tested, and methods to implement the test (Ahamed, 2023).* **This document will be created during the Program Design phase of development, as per Royce's model (Royce, 1970).**

3.9.4. Deadline

- *5 months after project start.*

3.10. Test Plan Specifications - Test Results

3.10.1. Author

- Technical Writer, Project Manager, Testing Team Lead, Testing Team Members

3.10.2. Intended Audience

- Testing Team Members, product clientele, future Operations engineers

3.10.3. Purpose

- The **Test Plan specifications document** will be updated to provide a history of all tests performed on the software and their results. It will accurately describe what tests have and have not been performed on the product. **This document will be created during the Testing phase of development, as per Royce's model (Royce, 1970).**

3.10.4. Deadline

- *15 months after the project starts.*

3.11. Operating Instructions

3.11.1. Author

- Technical Writer, Project Manager, Testing Team Lead, Coding Team Lead

3.11.2. Intended Audience

- Product clientele, future Operations engineers

3.11.3. Purpose

- The **Operating Instructions** will be utilized by the project clientele and future engineers to maintain and utilize the software. *It is an informational document that presents users with instructions on how to use the software (Wills, 2024).* **This document will be created during the Operations phase of development, as per Royce's model (Royce, 1970).**

3.11.4. Deadline

- *17 months after the project starts.*

4 ROLES

4.1. Project Manager (1 person)

4.1.1. Who

- Oliver

4.1.2. Qualifications

- Significant experience in development, testing, and management.

4.1.3. Responsibilities

- The project manager should manage both the development of the product and client relations. *They should also assist in drafting documentation and designing the finished product. The Project Manager will preside over the entire development process and aim the project towards completion (Ovcharenko, 2024).*

4.2. UX Designers (2 people)

4.2.1. Who

- Ursula and Xavier.

4.2.2. Qualifications

- College degree in or related to graphic design or visual media. Experience in the design industry.

4.2.3. Responsibilities

- The UX Designers will handle all design work for the entire project. They will create wireframes and mockups using design principles. They will also ensure the end-user portions of the product are visually pleasing. *In other words, they are responsible for the visual design of the software (Ovcharenko, 2024).*

4.3. Technical Writer (1 person)

4.3.1. Who

- Teri.

4.3.2. Qualifications

- College degree in or related to Technical Writing, Communications, or English.

4.3.3. Responsibilities

- The Technical Writer will create all of the text in the product. They are responsible for, alongside the coding team, creating and proofreading all text that will be displayed to the user. The Technical Writer will also proofread code comments for accuracy. Most importantly, the Technical Writer will draft, generate, edit, and proofread all documentation and required documents. They are responsible for high-quality, readable documentation.

4.4. Coding Team Lead (1 person)

4.4.1. Who

- Abe

4.4.2. Qualifications

- Significant experience in software engineering and project development.

4.4.3. Responsibilities

- The Coding Team Lead is responsible for overseeing the coding team and managing product development. They will designate tasks for team members and lead team meetings. Additionally, they will work with the Technical Writer to create required documents for the project.

4.5. Coding Team Members (6 people)

4.5.1. Who

- Britney, Claire, Grace, Ingrid, Jack, Larry

4.5.2. Qualifications

- College degree in software engineering, computer science, or a related field.

4.5.3. Responsibilities

- The Coding Team Member is responsible for developing the code for the project. They will collaborate with one another to create a working product.

4.6. Testing Team Lead (1 person)

4.6.1. Who

- Frank

4.6.2. Qualifications

- Significant experience in software debugging and testing.

4.6.3. Responsibilities

- The Testing Team Lead is responsible for overseeing the testing team and managing the debugging process. They will delegate tasks to team members and lead team meetings. Additionally, they will work with the Technical Writer to create required documents for the project.

4.7. Testing Team Member (4 people)

4.7.1. Who

- Emily, Frank, Holly, Keith

4.7.2. Qualifications

- College degree in software engineering, computer science, or a related field.

4.7.3. Responsibilities

- Testing Team Members will test and debug the code of the product, as developed by the coding team. They will collaborate with themselves and the Testing Team Lead to fully test the code for security flaws and bugs.

5 CHECKPOINTS

5.1. Overview Document Completed

5.1.1. Effort Estimation

- *2 weeks after project start.*

5.1.2. Exit Criteria

- *An Overview document will be created for every employee to read and understand. This checkpoint will only be completed upon creation of the Overview document.*

5.2. System Requirements Document Completed

5.2.1. Effort Estimation

- *1 month after the project starts.*

5.2.2. Exit Criteria

- *All system requirements should be identified. A System Requirements document will be created, showing all necessary system requirements. Though this document isn't shown in Royce's original model, documenting system requirements in an official document will assist in generation of the Software Requirements document and documentation of the project as a whole.*

5.3. Software Requirements Specification Completed

5.3.1. Effort Estimation

- *1.5 months after project start.*

5.3.2. Exit Criteria

- **All software requirements should be identified. A Software Requirements document will be created, showing all necessary software requirements. This checkpoint will only be completed upon creation of the Software Requirements document.**

5.4. Preliminary Design Specification Document Completed

5.4.1. Effort Estimation

- *2 months after the project starts.*

5.4.2. Exit Criteria

- **Before analysis, programming, and other work can be completed, a Preliminary Design Specification document must be created. This will outline a basic, introductory design of the software. This checkpoint will only be completed upon creation of the Preliminary Design Specifications document.**

5.5. Initial Software Documents Reviewed

5.5.1. Effort Estimation

- *2.5 months after project start.*

5.5.2. Exit Criteria

- **All software requirements and preliminary design documents should be reviewed by the client. This checkpoint will only be completed upon successful review of the Software Requirements document and Preliminary Design Specifications document by B.A.N.K.** *As such, this checkpoint should be developing during the generation of the Software Requirements and Preliminary Design document.*

5.6. Analysis Completed

5.6.1. Effort Estimation

- *3 months after project start.*

5.6.2. Exit Criteria

- *According to Paul Kirvan, the analysis stage requires an audit of system specifications, finances, and technical resources (Kirvan, 2024)*. **This checkpoint will only be completed upon full analysis of these specifications and requirements.**

5.7. Interface Design Specifications Document Completed

5.7.1. Effort Estimation

- *3.5 months after project start.*

5.7.2. Exit Criteria

- *The Interface Design Specifications document describes the different interfaces and their functionalities within the product (Arnab, 2015).* **This checkpoint will only be completed upon creation of the Interface Design Specifications document.**

5.8. Final Design Specifications Document - First Draft

5.8.1. Effort Estimation

- *4 months after project start.*

5.8.2. Exit Criteria

- **Dr. Royce's model requires several Final Design Specifications documents to be created and completed (Royce 1970). This checkpoint will only be completed upon creation of the first draft of the Final Design Specification document.**

5.9. Critical Software Review (CSR) #1

5.9.1. Effort Estimation

- *4 months after project start*

5.9.2. Exit Criteria

- **The purpose of a CSR meeting is to involve the client and allow them to review documentation and planning of the product. B.A.N.K. will review the Interface Design Specifications and Final Design First Draft documents. This checkpoint will only be completed upon a successful first CSR meeting.**

5.10. Final Design Specifications Document - Revised Draft Complete

5.10.1. Effort Estimation

- *4.5 months after project start*

5.10.2. Exit Criteria

- **After the first CSR Meeting, the Final Design will need to be rewritten. This checkpoint will only be completed upon creation of an Accepted Draft of the Final Design Specifications Document.**

5.11. Critical Software Review (CSR) #2

5.11.1. Effort Estimation

- *4.5 months after project start*

5.11.2. Exit Criteria

- **The purpose of a CSR meeting is to involve the client and allow them to review documentation and planning of the product. B.A.N.K. will review the Final Design Revised Draft. This checkpoint will only be completed upon a successful first CSR meeting.**

5.12. Test Plan Specifications Document

5.12.1. Effort Estimation

- *5 months after project start*

5.12.2. Exit Criteria

- **After the Final Design has been revised, a Test Plan Specifications document must be created, outlining potential tests to be performed on the software. This checkpoint will only be completed upon a successful first CSR meeting and creation of a Test Plan Specifications document.**

5.13. Electricity Usage Monitoring Coding

5.13.1. Effort Estimation

- *6 months after project start*

5.13.2. Exit Criteria

- **Users will be able to view and monitor their property's electricity usage. Pages must follow UX design as outlined in the Interface Specifications document. This is a milestone checkpoint, based on the completion of a major project milestone (Cser, 2023).**

5.14. Electricity Monitoring Coding

5.14.1. Effort Estimation

- *7.5 months after project start*

5.14.2. Exit Criteria

- **Users will be able to view their appliances and monitor their status. Maintenance tasks will be placed in a checklist, and the user will be able to check off items when they are complete. Pages must follow UX design as outlined in the Interface Specifications document.**

5.15. Home Maintenance Monitoring Coding

5.15.1. Effort Estimation

- *8.5 months after project start*

5.15.2. Exit Criteria

- **The application will provide a monthly checklist of home maintenance activities to be completed, and the user will be able to check off items when they are complete. Pages must follow UX design as outlined in the Interface Specifications document.**

5.16. Yard Maintenance Monitoring Coding

5.16.1. Effort Estimation

- *9.5 months after project start*

5.16.2. Exit Criteria

- **The software will provide a monthly checklist of home maintenance activities to be completed, and the user will be able to check off items when they are complete. Pages must follow UX design as outlined in the Interface Specifications document.**

5.17. Monthly Report Coding

5.17.1. Effort Estimation

- *10.5 months after project start*

5.17.2. Exit Criteria

- **The software will generate a report at the end of every month, including a summary of recent monitoring and completed tasks. Pages must follow UX design as outlined in the Interface Specifications document.**

5.18. Module Integration

5.18.1. Effort Estimation

- *12 months after project start*

5.18.2. Exit Criteria

- **This is an integration checkpoint. As such, it will ensure that different components of the software work together as intended (Cser, 2023). All monitoring and checklist software will work together as intended. All software will be fully compatible with one another and work as one application.**

5.19. Final Design Specifications (As Built)

5.19.1. Effort Estimation

- *12.5 months after project start*

5.19.2. Exit Criteria

- **This is the final version of the Final Design document. It will contain a complete description of the software. This checkpoint will only be completed upon creation of the Final Design Specifications (As Built) document.**

5.20. Testing Complete

5.20.1. Effort Estimation

- *15 months after project start*

5.20.2. Exit Criteria

- **All application features have been tested and approved. The system has been thoroughly debugged and follows all security and safety guidelines. Additionally, the Test Plan document is updated to contain a comprehensive record of tests performed - becoming the Test Plan .**

5.21. Final Acceptance Review (FSAR)

5.21.1. Effort Estimation

- *16 months after project start*

5.21.2. Exit Criteria

- **The client, B.A.N.K., will thoroughly inspect, test, and review the completed software and documentation. All documents and software must be approved by the client for production.**

5.22. Operating Instructions Complete

5.22.1. Effort Estimation

- *17 months after project start*

5.22.2. Exit Criteria

- **Documentation for operation of the software must be completed for effective maintenance (Royce, 1970). This checkpoint will only be completed upon creation of the Operating Instructions Specifications document.**

5.23. Operations Deployed

5.23.1. Effort Estimation

- *17 months after project start*

5.23.2. Exit Criteria

- **With all documentation, development, and debugging complete, the product is ready for deployment. This checkpoint will only be completed upon successful deployment of the product.**

5.24. Timeline

Event	Date
Overview Document Completed Deliverable: Overview Document	2 weeks after project start.
System Requirements Document Completed Deliverable: System Requirements Document	1 month after project start.
Software Requirements Specification Completed Deliverable: Software Requirements Document	1.5 months after project start.
Preliminary Design Specification Document Completed Deliverable: Preliminary Design Specification Document	2 months after project start.
Initial Software Documents Reviewed	2.5 months after project start.
Analysis Completed	3 months after project start.
Interface Design Specifications Document Completed	3.5 months after project start.

Deliverable: Interface Design Specification Document	
Final Design Specifications Document - First Draft Deliverable: Specifications Document - First Draft	4 months after project start.
Critical Software Review (CSR) #1	4 months after project start.
Final Design Specifications Document - Revised Draft Complete Deliverable: Final Design Specifications Document - Revised Draft	4.5 months after project start.
Critical Software Review (CSR) #2	4.5 months after project start.
Test Plan Specifications Document Complete Deliverable: Preliminary Design Specification Document	5 months after project start.
Electricity Usage Monitoring Coding Complete Deliverable: Electricity Usage Monitoring Module	6 months after project start.
Electricity Monitoring Coding Complete	7.5 months after project start.

Deliverable: Electricity Monitoring Module	
Home Maintenance Monitoring Coding Deliverable: Home Maintenance Module	8.5 months after project start.
Yard Maintenance Monitoring Coding Deliverable: Yard Maintenance Monitoring Module	9.5 months after project start.
Monthly Report Coding Deliverable: Monthly Report Module	10.5 months after project start.
Module Integration Deliverable: Software with Integrated Modules	12 months after project start.
Final Design Specifications Document (As Built) Complete Deliverable: Final Design Specifications (As Built) Document	12.5 months after project start.
Testing Complete Deliverable: Test Plan Specifications - Test Results	15 months after project start.
Final Acceptance Review (FSAR)	16 months after project start.

Operating Instructions Complete <u>Deliverable: Operating Instructions</u>	17 months after project start.
Operations Deployed <u>Deliverable: Finished Product</u>	17 months after project start.

6 ANALYSIS AND REFLECTION

As introduced by Royce in his paper, the Waterfall approach presents several advantages to the Software Development endeavour, but it also suffers from some disadvantages. In a general view, we can consider those in the following table (Shravan, 2023).

Table 1

Advantages and disadvantages of the Waterfall Model for Software Development

Advantages/Strengths	Disadvantages/Weaknesses
• Structured and predictable. • Comprehensive documentation. • Easy management. • Clear deliverables. • Suitable for small projects.	• Limited flexibility. • Lack of customer involvement. • High risk. • Lengthy development cycle. • Not suitable for complex projects

We can see that for the ends of our project, the model is advantageous for the most part. Considering the budget, time and personnel we have at hand, the possible advantages of the model become a strength, and the possible disadvantages are reduced.

Now, under a specific view of our project, and under the consideration of the *Overview* section of this paper, there may be some elements that we can consider not entirely necessary, and they could be subject to reduction.

We mentioned that the basic version of the Waterfall Model (*Figure 1*) was enhanced by the addition of extras features (*Figure 2*). We understand that the complete implementation of those additional steps may not be beneficial for our particular work, considering the size of the project, the real needs of the customer and the human resources we have at hand. Also, we need to consider that the state of art is different today than it was at the time of Royce's paper, and some of the problems he was considering during that time may have become a trivial problem for modern software developers.

Going back to the main features (or phases) of the model, as explained in the *Overview* section text and in Figure 2, we will comment on each of the 5 additional features (or phases) and the possible work reductions we could face (if any).

1) Extensive design:
    a) This is a necessary phase that should be fully implemented.
2) Complete documentation:
    a) Royce (1970) considers that documentation is the first most important criterion for success around product originality.
    b) Though documentation is an important artifact of the development process, it may not be necessary to involve our customer to details below the level of UX/UI. So it is possible that during the design time we decide to omit some information from the customer, information that possibly has less value for them than for us.
    c) For internal use, for the most part, we may want to limit the production of documentation for the obscurest part of the system and for future product maintainers.
3) Pilot model:
    a) Second to documentation in terms of criterion of success (Royce, 1970).
    b) For this project we have decided to omit the pilot model implementation, under the basis of the low complexity of the system. We may decide later

in time to change this decision, once we are able to advance through the design phase.

4) Plan, control and monitor testing:
    a) Testing is an extension of previous phases, and it aims to uncover and then solve problems that those phases were not able to detect and eliminate.
    b) We consider that the approach to testing proposed by Royce is reasonable and between our means of budget and time constraints.
5) Involve the customer:
    a) This feature is common practice in modern day software development, and is not an impediment to our project.

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